

9.1. Warm-up 9

9.1. Eq. of the circle?

a) diam [A,B], A(1,2), B(-3,1)

$$\boxed{C: (x-h)^2 + (y-k)^2 = R^2} \text{ CIRCLE.}$$

$C(h,k)$ = midpoint of diameter AB.

$$h = \frac{x_A + x_B}{2} = \frac{1-3}{2} = -1 \Rightarrow C(-1, \frac{1}{2})$$

$$k = \frac{y_A + y_B}{2} = \frac{1}{2}$$

$$AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2} = \sqrt{(-4)^2 + (-1-2)^2} = \sqrt{16+9} = 5. \Rightarrow R = \frac{AB}{2} = \frac{5}{2}.$$

$$\Rightarrow C: (x+1)^2 + (y-\frac{1}{2})^2 = (\frac{5}{2})^2.$$

b) center I(2,-3), R=7.

$$\Rightarrow C: (x-2)^2 + (y+3)^2 = 7^2$$

c) center I(-1,2), passing through A(2,6).

$$A \in C \Rightarrow R = \sqrt{(x_A - x_I)^2 + (y_A - y_I)^2} = \sqrt{(2+1)^2 + (6-2)^2} = \sqrt{9+16} = 5.$$

$$\Rightarrow C: (x+1)^2 + (y-2)^2 = 5^2.$$

d) centered at O(0,0), tangent to l: $3x - 4y + 20 = 0$.

$$\boxed{\text{radius} \perp \text{tg}} \Rightarrow d(O, l) = \frac{|3 \cdot 0 - 4 \cdot 0 + 20|}{\sqrt{3^2 + 4^2}} = \frac{20}{\sqrt{25}} = \frac{20}{5} = 4. = R.$$

$$\Rightarrow C: x^2 + y^2 = 4^2.$$

e) passing through A(3,1), B(-1,3) and having the center on the line

$$l: 3x - y - 2 = 0.$$

(H1) O = center.

$$O \in l$$

$O \in b$ (perpendicular bisector b of [AB]).

$$\Rightarrow O = l \cap b.$$

$$R = d(O, A) = d(O, B) \dots$$

(H2) $C: (x-x_0)^2 + (y-y_0)^2 = R^2 \rightarrow C(x_0, y_0)$ = center of C.

$$P \in d: 3x - y - 2 = 0 \text{ (diam. line)} \Rightarrow y_0 = 3x_0 - 2 \Rightarrow P(x_0, 3x_0 - 2)$$

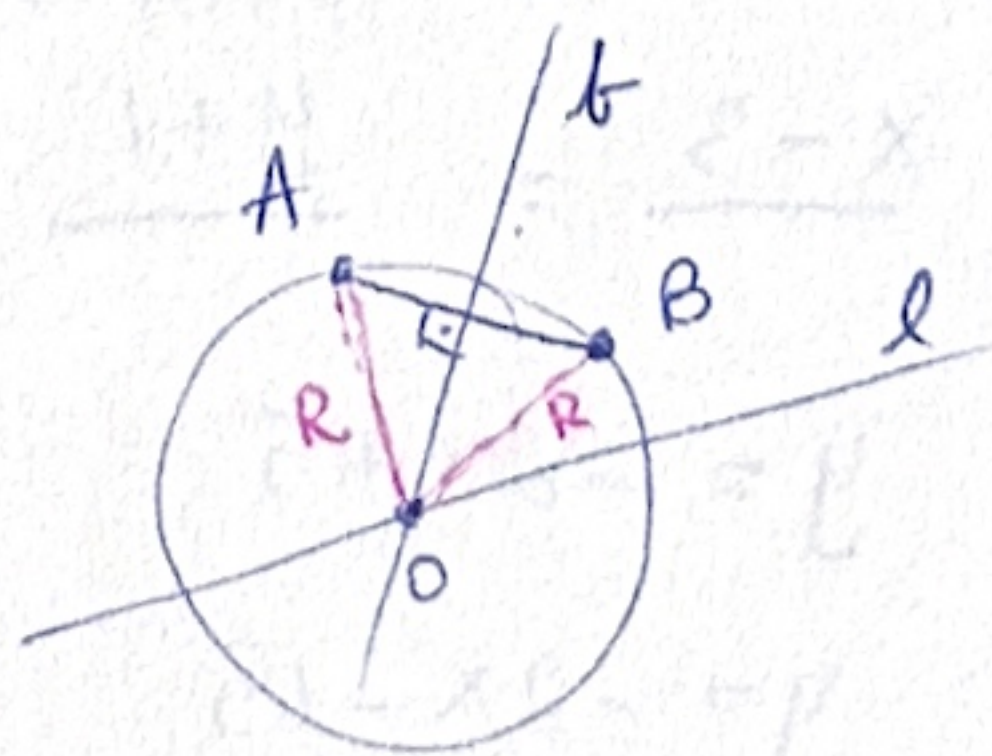
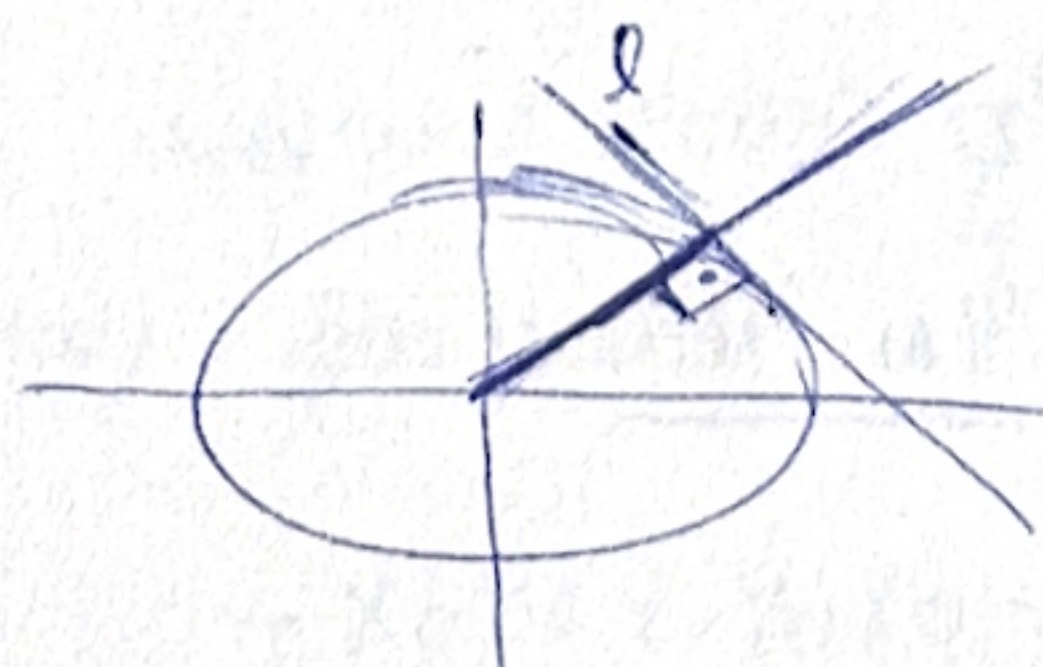
$$C: (x-x_0)^2 + (y-3x_0+2)^2 = R^2$$

$$A(3,1) \in C \Rightarrow (3-x_0)^2 + (1-3x_0+2)^2 = R^2 \Rightarrow \dots \Rightarrow x_0 = 2$$

$$B(-1,3) \in C \Rightarrow (-1-y_0)^2 + (3-3x_0+2)^2 = R^2 \Rightarrow y_0 = 4.$$

$$\Rightarrow \text{center: } C(2,4)$$

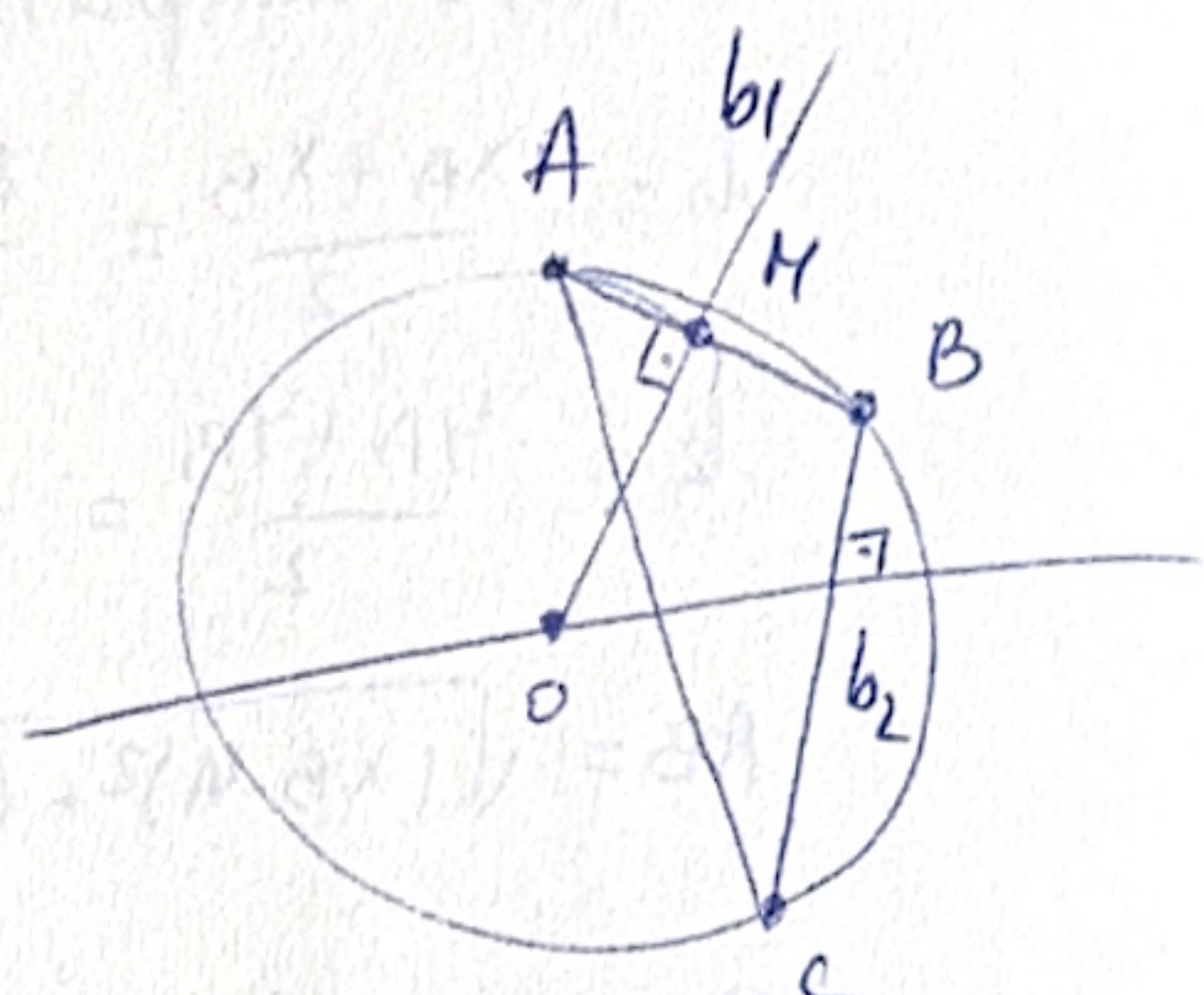
$$\text{radius: } R = d(C, A) = d(C, B) = \dots = \sqrt{10}.$$



f) passing through $A(1,1), B(1,-1), C(2,0)$ $\Rightarrow \dots$

(K1) : eg:
$$\begin{vmatrix} x^2+y^2 & x & y & 1 \\ x_A^2+y_A^2 & x_A & y_A & 1 \\ x_B^2+y_B^2 & x_B & y_B & 1 \\ x_C^2+y_C^2 & x_C & y_C & 1 \end{vmatrix} = 0.$$

(K2) $O \in b_1 \rightarrow [AB]$
 $O \in b_2 \rightarrow [BC]$



$O = b_1 \cap b_2$

$r = d(O, A)$

$b_1: x_M = \frac{x_B + x_A}{2} = \frac{2}{2} = 1, y_M = \dots = 0 \Rightarrow M(1,0)$

$m_{AB} = \frac{y_B - y_A}{x_B - x_A} = \frac{-1-1}{1-1} \Rightarrow \text{horizontal line} \Rightarrow b_1 = 0.$
 $\hookrightarrow \text{vertical}$

$b_2: x_N = 1.5, y_N = -0.5 \Rightarrow N\left(\frac{3}{2}, -\frac{1}{2}\right)$

$m_{BC} = 1 \Rightarrow m_{b_2} = -1 \Rightarrow b_2: y - \left(-\frac{1}{2}\right) = -1\left(x - \frac{3}{2}\right)$
 $\Rightarrow b_2: y = -x + 1.$

g) Ag to both $l_1: 2x + y - 5 = 0$ and $l_2: 2x + y + 15 = 0$ if one Ag. point is M

$\Rightarrow l_1 \parallel l_2$

$l: MN: \frac{x_N - x_M}{y_N - y_M} = \frac{y_N - y_M}{x_N - x_M}$

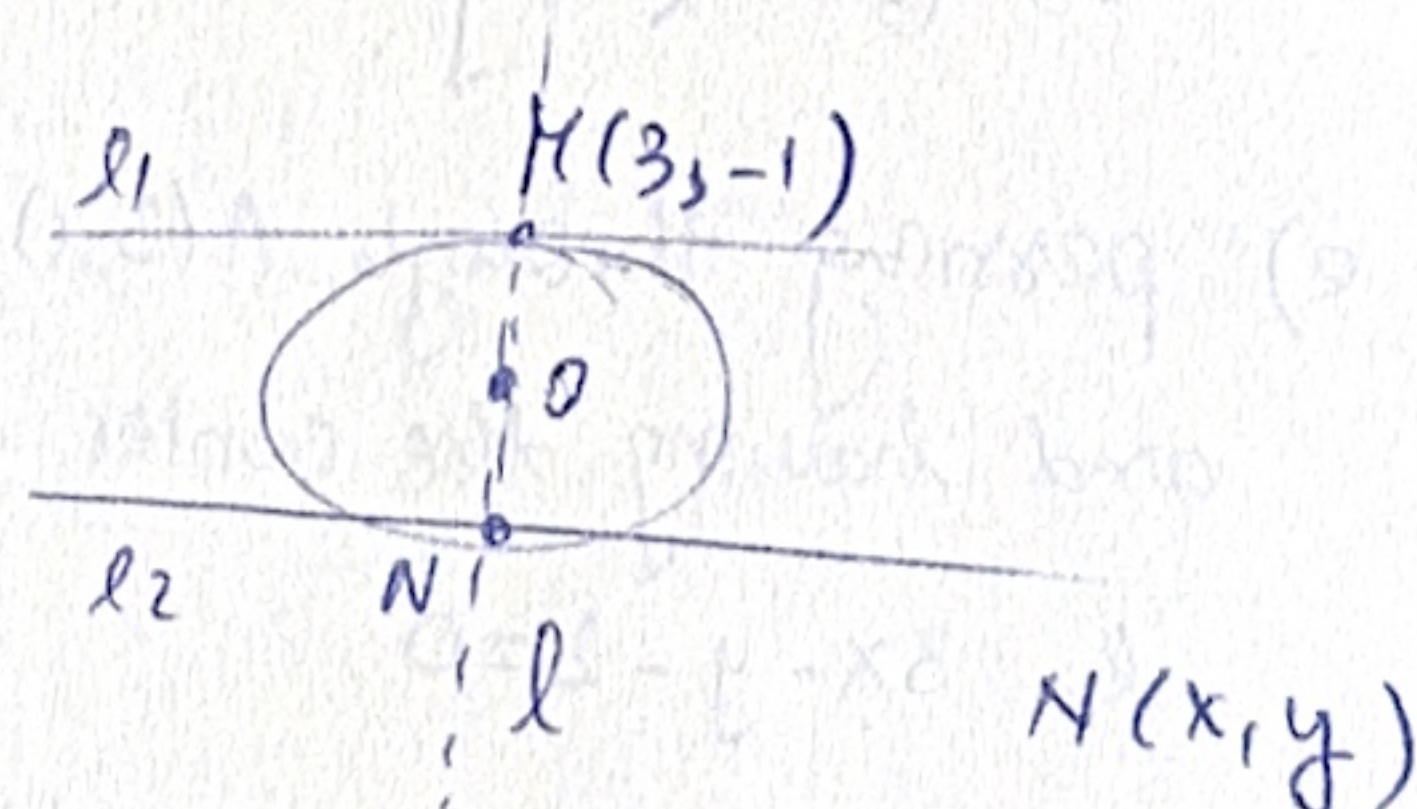
$l: MN: \frac{x-3}{y+1} = \frac{y+1}{x-3}$

$l_1: y = -2x + 5$

$l_2: y = -2x - 15$

$\Rightarrow m_{l_1} = m_{l_2} = -2 \quad (l_1 \parallel l_2)$

$\Rightarrow m_l = \frac{-1}{-2} = \frac{1}{2}$



$l: MN: y - (-1) = \frac{1}{2}(x - 3)$

$l: MN: 2(y+1) = x-3$

$l: MN: 2y+2 = x-3$

$l: MN: -x + 2y + 5 = 0.$

$N = l \cap l_2 \Rightarrow \begin{cases} -x + 2y + 5 = 0 \quad | \cdot 2 \Rightarrow \\ 2x + y + 15 = 0 \end{cases} \Rightarrow \begin{cases} -2x + 4y + 10 = 0 \\ 2x + y + 15 = 0 \end{cases} \quad (+)$

$O = \frac{x_M + x_N}{2} = \frac{3-5}{2} = -1, y_O = \frac{y_M + y_N}{2} = \frac{-1-5}{2} = -3.$

$5x + 25 = 0$

$x = -5.$

$\Rightarrow O(-1, -3); r = d(O, M) = \dots = \sqrt{20}$
 $\Rightarrow \text{Eq: } (x+1)^2 + (y+3)^2 = 20; y = -5 \Rightarrow M(-5, -5)$

9.2. $\mathcal{C}$ of radius R .

a) param. $x \mapsto (x, \pm \sqrt{R^2 - x^2}) \Rightarrow$ param. of tg. lines to $\mathcal{C}$.

$$f(x) = (x, \sqrt{R^2 - x^2}), \quad f'(x) = \left(1, -\frac{x}{\sqrt{R^2 - x^2}}\right)$$

l = tg line to $\mathcal{C}$ with contact point $f(x_0) \in \mathcal{C}$.

$$l: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_0 \\ \sqrt{R^2 - x_0^2} \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ -\frac{x_0}{\sqrt{R^2 - x_0^2}} \end{bmatrix}, \quad t \in \mathbb{R}.$$

$$! \quad \langle f(x), f'(x) \rangle = f(x) \cdot f'(x) = x \cdot 1 - \sqrt{R^2 - x^2} \cdot \frac{x}{\sqrt{R^2 - x^2}} = x - x = 0$$

$\Rightarrow$ pos. vector of $f(x_0) \in \mathcal{C}$

$\perp$

dir. vector $f'(x_0)$.

b) param. $\theta \mapsto \underbrace{(R \cos \theta, R \sin \theta)}_{f(\theta)} \Rightarrow$ param. of tg. lines to $\mathcal{C}$.

$$f'(\theta) = (-R \sin \theta, R \cos \theta)$$

$$l: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} R \cos \theta_0 \\ R \sin \theta_0 \end{bmatrix} + t \cdot \begin{bmatrix} -R \sin \theta_0 \\ R \cos \theta_0 \end{bmatrix}$$

$f(\theta) \perp f'(\theta)$
 $\downarrow \quad \quad \downarrow$
 pos. vector $\perp$ dir. vector

c) compare with $x x_0 + y y_0 = R^2, \quad (x_0, y_0) \in \mathcal{C}$.

$\hookrightarrow m = (x_0, y_0)$ normal vector

$\hookrightarrow v = (-y_0, x_0)$ pos. vector

$$m \cdot v = 0$$

$$-x_0 y_0 + y_0 x_0 = 0$$

usually normal direction = 0,

but here normal position = 0

v orthogonal to m

If $f(x_0) = (x_0, y_0) \Rightarrow$ a).

If $f(\theta_0) = (x_0, y_0) \Rightarrow$ b).

9.3. $l: \mathcal{E}: \begin{cases} x^2 + 3y^2 - 25 = 0 \\ x + 2y - 7 = 0 \Rightarrow x = 7 - 2y \end{cases}$

$$\Rightarrow (7 - 2y)^2 + 3y^2 - 25 = 0$$

$$\dots \Rightarrow y = 2 \pm \frac{2}{\sqrt{7}}$$

$$\Rightarrow A(7 - 2y_1, y_1), \quad B(7 - 2y_2, y_2)$$

$\mathcal{E}: x^2 + 3y^2 - 25 = 0$ what surface
 $\Delta = 0 - 4 \cdot 1 \cdot (-25) = 100$

9.4. $\ln \mathcal{H} \begin{cases} \dots \uparrow \\ \dots \end{cases} \Rightarrow 2 \text{ points.}$

9.5. tangents to $\mathcal{H}$: $\frac{x^2}{16} - \frac{y^2}{8} - 1 = 0$ which are ll: $4x + 2y - 5 = 0$.

$$\boxed{\text{Tangents of hyperbola: } y = kx \pm \sqrt{a^2 k^2 - b^2} \mid \text{ if } k \in (-\infty, -\frac{b}{a}) \cup (\frac{b}{a}, \infty)}$$

$$a=4, b=2\sqrt{2}, k=-2.$$

$$l: 2y = 5 - 4x$$

$$l: y = -\frac{4}{2}x + \frac{5}{2}$$

$$l: y = -2x + \frac{5}{2} \Rightarrow \text{M}l = -2.$$

$$l_1: y = -2x + \sqrt{16 \cdot 4 - 8} = -2x + 2\sqrt{14}$$

$$l_2: y = -2x - 2\sqrt{14}.$$

9.6 tangents to $\mathcal{H}$: $x^2 - y^2 = 16$ which contain the point $M(-1, 7)$

$$l: \boxed{y = kx \pm \sqrt{a^2 k^2 - b^2}} \mid k \in (-\infty, -\frac{b}{a}) \cup (\frac{b}{a}, \infty)$$

$$\mathcal{H}: x^2 - y^2 = 16 \mid :16$$

$$\mathcal{H}: \frac{x^2}{16} - \frac{y^2}{16} = 1. \Rightarrow a=b=4.$$

$$M(-1, 7) \in l \text{ and } a=b=4.$$

$$l: y = -k \pm \sqrt{16k^2 - 16}$$

$$l: 7 + k = \pm \sqrt{16(k^2 - 1)} \mid ^2$$

$$l: 49 + 14k + k^2 = 16k^2 - 16 \quad (k^2 \geq \frac{b^2}{a^2})$$

$$\Rightarrow l: 15k^2 - 14k - 65 = 0$$

$$\Delta = 2^4 \cdot 12$$

$$k_{1,2} = \begin{cases} k_1 = -\frac{5}{3} \\ k_2 = \frac{13}{5} \end{cases} \Rightarrow$$

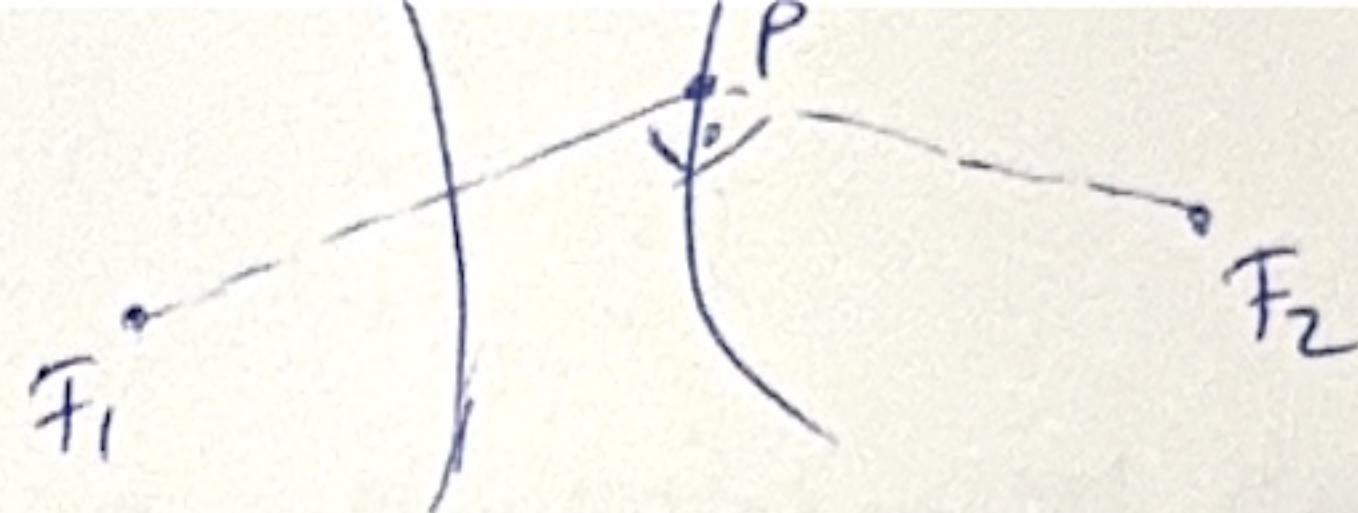
$$y_1 = -\frac{5}{3}x + \dots$$

$$y_2 = \frac{13}{5}x + \dots$$

9.7. $\mathcal{H}: x^2 - \frac{y^2}{4} - 1 = 0$

F_1, F_2 foci.

$M \in \mathcal{H}$ s.t. $\angle F_1 M F_2 = 90^\circ$



$\Rightarrow$ the points are the $\cap$ of $\mathcal{H}$ with a circle centered at O and passing through F_1, F_2 .

9.8. $k = ?$ s.t. $\underbrace{y = kx + 2}_{l_k}$ tangent to $\underbrace{P: y^2 = 4x}_{P_2}$

$l_k \cap P_2: \begin{cases} y^2 = 4x \\ y = kx + 2 \end{cases} \Rightarrow (kx + 2)^2 = 4x$
 $\Rightarrow \dots \Rightarrow \Delta = 4^2(-2k - 1)$

l_k tg to P_2 if $\Delta = 0 \Leftrightarrow k = -\frac{1}{2}$.

9.9. $P: y^2 = 16x$; tangents to P which are: $\Rightarrow \boxed{p=8}$

a) $\parallel$ to $l: 3x - 2y + 30 = 0 \Rightarrow 2y = 3x + 30$

$y = \frac{3}{2}x + 15 \Rightarrow m = \frac{3}{2} = k$

b) $\perp$ to $l: 4x + 2y + 7 = 0.$

$\Rightarrow 2y = -7 - 4x$

$y = -2x - \frac{7}{2} \Rightarrow k = \frac{-1}{2} = \frac{1}{2}$

$\boxed{y = kx + \frac{p}{2k}}$ for tg.

a) $y = \frac{3}{2}x + \frac{8}{2 \cdot \frac{3}{2}} = \frac{3}{2}x + \frac{8}{3}$

b) $y = \frac{1}{2}x + \frac{8}{2 \cdot \frac{1}{2}} = \frac{1}{2}x + 8$

9.10. tg to $P: y^2 = 16x$, $P(-2, 2) \in \text{tg}$. $\boxed{p=8}$

$\boxed{y = kx + \frac{p}{2k}} = kx + \frac{8}{2k} = kx + \frac{4k}{k} = \cancel{k(x+4)}$

~~$2 = k(-2+4) \Rightarrow 2 = 2k \Rightarrow k=1.$~~

$2 = -2k + \frac{4}{k} \mid \cdot k \Rightarrow 2k = -2k^2 + 4$

$\Rightarrow 2k^2 + 2k - 4 = 0 \Rightarrow k^2 + k - 2 = 0$

$\Delta = 1 - 4 \cdot (-2) = 9$

$l_1: y = -2x + \frac{8}{2 \cdot (-2)} = -2x - 2$

$k_{1,2} = \frac{-1 \pm 3}{2} \rightarrow k_1 = -2$
 $\rightarrow k_2 = 1$

$l_2: y = x + \frac{8}{2} = x + 4.$